

Quantifying Finfluencer Influence on Generation Z Investment Decisions using Multiple Linear Regression

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Abstract—The democratization of financial markets through social networks and zero-commission platforms has introduced significant socio-economic risks for Generation Z retail investors. While traditional behavioral finance research focuses on macro-level market sentiment, there remains a critical gap in quantifying how individual trust in digital influencers interacts with foundational literacy and experience to drive risky trading. We propose a micro-behavioral framework utilizing Multiple Linear Regression (MLR) to isolate the impact of "finfluencers" on individual investment execution. Using an augmented dataset ($N = 546$), we tested the influence of Social Media Exposure (SME), Influencer Credibility and Trust (ICT), Herding Behavior (HB), and Overconfidence (OC), while controlling for Financial Literacy (FL) and Investment Experience (IE). Statistical analysis confirms the model is highly significant ($p < 0.001$) and explains 64.5% of the variance in aggressive behavior ($R^2 = 0.645$). Results identify ICT ($\beta = 0.5953$) and SME ($\beta = 0.1927$) as primary positive drivers, while IE ($\beta = -0.4530$) serves as a crucial protective factor against impulsive execution. The model achieved a 59.56% improvement in predictive accuracy over baseline methods, validating the necessity of micro-behavioral metrics in modern retail risk profiling and regulatory interventions.

Index Terms—Finfluencers, Behavioral Finance, Retail Trading, Financial Literacy, Multiple Linear Regression, Sentiment Analysis.

I. INTRODUCTION

The global landscape of retail trading is currently undergoing a structural transformation of unprecedented scale. Driven by the proliferation of zero-commission brokerages, frictionless mobile interfaces, and the democratization of market access, a massive wave of Generation Z retail investors has entered the financial markets [1]. This shift is not merely demographic but cultural; the traditional barriers to entry—high capital requirements and institutional gatekeeping—have been replaced by a digital ecosystem that encourages immediate, high-frequency participation. However, instead of relying on fundamental analysis, professional equity research, or traditional wealth managers, this new cohort largely consumes financial intelligence through algorithmically driven social media platforms. This paradigm shift has given rise to a new class of "finfluencers" and unregulated content creators who now wield localized and systemic power over retail capital

allocation, often overriding market fundamentals with viral sentiment [2].

To understand this disruption, one must examine the underlying mechanics of modern retail trading psychology. In classical behavioral finance, decisions are frequently distorted by cognitive biases such as overconfidence (OC) and herding behavior (HB). In the contemporary digital environment, these biases are significantly accelerated by the constructs of Influencer Credibility and Trust (ICT) and Social Media Exposure (SME) [3]. Unlike traditional financial advisors whose authority is derived from institutional credentials, finfluencers often build "parasocial" bonds with their audience, where perceived authenticity and lifestyle alignment are mistaken for financial expertise. Traditionally, the primary defense against such psychological distortions is formal financial literacy (FL), which serves as the baseline knowledge required to evaluate risk-return profiles accurately. The central conflict now facing regulators and economists is whether algorithmic digital nudges and parasocial trust can statistically override this foundational literacy to trigger Aggressive Investor Behavior (AIB) [4].

Current research attempts to monitor this structural disruption primarily from a macro-level perspective, focusing on aggregate market movements rather than individual decision-making. The dominant methodology utilizes Natural Language Processing (NLP) and advanced deep learning architectures to conduct massive-scale sentiment analysis on millions of social media posts, aiming to predict broad asset volatility and "meme stock" rallies [5]. Concurrently, researchers frequently employ hierarchical regression models to map how general cognitive biases drive risky trading decisions within emerging markets, particularly among younger cohorts [6]. While these methodologies provide an essential view of market-wide dynamics, they possess a significant blind spot: they often overlook the nuanced, micro-level interactions between an individual user and the specific influencers they trust.

Despite the sophistication of these models, the current body of literature suffers from three critical limitations. First, algorithmic sentiment models tend to treat retail traders as a monolithic entity, effectively ignoring the diverse micro-

level psychology and varying experience levels of individual investors [7]. Second, existing behavioral models frequently fail to isolate and quantify the specific degree of trust users place in individual finfluencers as a distinct variable from general social media usage. Finally, these predictive models routinely lack baseline financial literacy as a control variable, implicitly assuming that viral social media hype manipulates all retail investors with equal efficacy [8]. When analytical models cannot mathematically distinguish between a fundamentally sound, data-driven trade and a blind gamble based on digital credibility, retail risk is fundamentally mispriced.

To address these empirical gaps, this paper proposes a quantitative, micro-behavioral framework designed to mathematically isolate the impact of finfluencers on individual trading execution [9]. Utilizing an Ordinary Least Squares (OLS) regression framework on a primary dataset of 546 verified Generation Z investors, this research tests a multi-variable hypothesis to determine how ICT, SME, and Investment Experience (IE) drive aggressive execution. Crucially, the methodology integrates formal Financial Literacy as a strict control variable, providing a statistical test of whether digital sentiment can override actual financial knowledge in a high-velocity environment [10]. By identifying the "protective" role of experience against the "catalytic" role of digital trust, this study delivers the integrated, quantitative evaluation of the modern retail investor that existing sentiment systems currently lack.

II. LITERATURE REVIEW

The recent evolution of retail trading is characterized by a paradigm shift where traditional financial advisory models are being challenged by digital, algorithmically driven content. This literature review evaluates the existing research across four key dimensions: advanced predictive modeling, the socio-psychological mechanics of digital influence, the impact of online sentiment on specific asset classes, and the demographic factors shaping modern investment behavior. The transition from institutional guidance to decentralized social information requires a robust understanding of both the technical and behavioral frameworks governing the modern investor.

A. Behavioral Biases and Digital Nudges

The psychological landscape of modern investment is increasingly defined by the interface between human cognition and digital platforms. Experimental investigations into gamified risk-taking indicate that digital interface designs can fundamentally alter an investor's risk appetite, often triggering more aggressive execution strategies than would be observed in traditional settings [12]. This shift is not merely a byproduct of technology but is actively managed through strategic creator roles; research into finfluencer engagement suggests that while these creators often position themselves as educational advocates, they operate within complex strategic partnership ecosystems that prioritize community engagement over objective financial analysis [13].

The consequences of this engagement are particularly visible in high-activity digital communities. Evidence from the r/WallStreetBets ecosystem confirms that concentrated social media attention frequently leads to uninformed trading execution, resulting in a higher probability of realizing negative returns compared to traditional market strategies [14]. This phenomenon is exacerbated by the professionalization of influencer marketing, which has emerged as a primary force in shaping the investment preferences and execution strategies of retail investors in the digital age [15]. Consequently, the "nudge" effect of social media acts as a catalyst for behavioral biases that override fundamental market logic.

B. Institutional Transitions and Quality Frameworks

The regional context of digital reliance plays a significant role in determining how these biases manifest in emerging markets. In the Indian financial landscape, retail investment perceptions are increasingly shaped by a heavy reliance on social media for real-time market insights, which frequently correlates with heightened herding behavior and a decrease in independent fundamental research [19]. To address these risks, technical frameworks have been proposed to integrate non-traditional data into financial forecasting; for instance, deep-learning-based stock market predictions have begun incorporating Environmental, Social, and Governance (ESG) sentiment alongside technical indicators to achieve a more holistic view of asset movement [11].

Despite these technical advancements, a significant psychological barrier remains centered on the individual's self-perception. Researchers have identified profound financial self-confidence gaps that emerge when young consumers rely on digital information sources, noting that gendered differences in risk appetite and confidence become more pronounced in algorithmically driven environments [20]. To mitigate these disparities, multidimensional quality frameworks are being developed to help retail investors and institutional partners navigate the rise of finfluencers by ensuring that the quality of digital advice meets professional standards [21]. These frameworks aim to establish institutional accountability in a decentralized information ecosystem.

C. Algorithmic Architectures and Market Sentiment

The role of digital creators extends beyond simple information dissemination, representing a broader financialization of digital life. Critics of the finfluence phenomenon suggest that influencers are not merely market participants but are actively participating in economic meaning-making for a generation that views traditional institutional finance as fundamentally inaccessible [22]. This transition has necessitated the development of more granular analytical tools; by using the dynamic semantic structure of digital news flow, researchers have demonstrated that the structural patterns of information delivery on platforms like Twitter are as predictive of market movement as the actual content of the news [16].

The predictive power of these digital behaviors is further refined through the analysis of high-profile social signals.

TABLE I
LITERATURE REVIEW SUMMARY

Ref	Objective	Domain	Methodology	Dataset	Findings	Limitations
[7]	Decipher social media sentiment effects on cryptocurrency market dynamics.	Cryptocurrency / Social Sentiment	Sentiment deciphering and market dynamics modeling.	Social media data and cryptocurrency market metrics.	Social media sentiment exerts a tangible impact on cryptocurrency price trends.	Scope is restricted to decentralized digital assets.
[11]	Prediction incorporating ESG sentiment and technical indicators.	Deep Learning / Market Forecasting	Deep-learning-based stock prediction models.	Stock market data and ESG sentiment scores.	Integrating ESG sentiment significantly improves model predictive accuracy.	Heavy reliance on automated sentiment labeling accuracy.
[12]	Analyze gamified digital nudges on risk-taking behavior.	Gamification / Behavioral Finance	Experimental design and simulation.	Retail investors in simulated platforms.	Gamification increases risk taking; formal literacy reduces the effect.	Experimental setting may differ from live market execution.
[13]	Shed light on influencers' motivations and role in investor engagement.	Finfluencers / Financial Ed.	Online survey; quantitative descriptive statistics.	Representative sample of German-speaking finfluencers.	Finfluencers see themselves as advocates; strategic partnerships influence role.	German-speaking context only; self-reported data subject to bias.
[14]	Study influence of Reddit attention on retail investors' risk-taking.	Social Sentiment / Market Dynamics	NLP sentiment scoring; holding period return analysis.	Individual trading data and WSB posts.	High attention spurs uninformed trading and realizes negative returns.	NDA limits replication; findings specific to Reddit culture.
[15]	Examine how influencer marketing shapes investment preferences via credibility.	Influencer Marketing / Digital Finance	Quantitative survey; source credibility theory analysis.	Retail investors active on social media platforms.	Influencer credibility and trustworthiness determine decisions.	Specific sample size not disclosed; does not control for financial literacy.
[16]	Using dynamic semantic structure of news flow to enhance forecasting.	Semantic Analysis / Forecasting	Twelve-year study on Twitter news channels.	Historical news flow from Twitter channels.	Structural patterns in news delivery are highly predictive of price shifts.	Limited to short-form news platforms like Twitter.
[17]	Stock market forecasting using neural networks and sentiment analysis.	Artificial Intelligence / Finance	Neural network modeling with multi-factor inputs.	Fundamental indicators and market sentiment.	Multi-factor neural networks provide superior forecasting in retail markets.	Requires high-frequency data for optimal training performance.
[18]	Enhance stock predictions by classifying unlabeled celebrity Twitter data.	Social Media / Market Prediction	Classification techniques for unlabeled celebrity data.	Unlabeled celebrity Twitter data and stock metrics.	Incorporating celebrity social data improves stock movement prediction accuracy.	Reliance on celebrity-specific discourse for market relevance.

TABLE II
COMPARISON OF PROPOSED METHODOLOGY WITH EXISTING LITERATURE

Ref.	Methodology	Micro-Behavior	Indiv. Trust	Literacy Control	Quant. Causality
[6], [7]	NLP / Deep Learning	×	×	×	×
[4], [10]	SEM / Behavioral	✓	×	✓	✓
[15], [19]	Qualitative / Survey	✓	✓	×	×
Proposed	MLR Model	✓	✓	✓	✓

Note: **Micro-Behavior** refers to the quantification of individual psychological triggers rather than macro-market sentiment; **Indiv. Trust** represents the specific measurement of Influencer Credibility and Trust (ICT); **Literacy Control** denotes the integration of formal Financial Literacy (FL) as a statistical baseline; **Quant. Causality** indicates the use of OLS regression to establish mathematical weights (β) for each predictor.

Recent work in stock market prediction has shown that classifying unlabelled data from celebrity Twitter accounts can effectively capture the ripple effects of high-profile social activity on market movements [18]. This platform-specific sentiment is highly sensitive to geopolitical and technological catalysts; for example, Reddit sentiment analysis reveals that public perception of conflicts or emerging technologies like the Metaverse can cause immediate and drastic shifts in the financial outlook of specific digital coins [23]. This reinforces the necessity of including platform-specific dynamics in any comprehensive behavioral framework.

D. Socio-Economic Predictions and Socialization

Managing the inherent volatility of retail markets requires the deployment of advanced neural network models. Current research suggests that stock market forecasting is significantly enhanced when neural networks synthesize fundamental indicators with market sentiment analysis, providing more stable forecasts in an increasingly irrational retail environment [17]. By fusing these technical indicators with sentiment analysis in hybrid deep learning frameworks, researchers can now predict stock price movements with greater accuracy by modeling the simultaneous impact of multi-source digital signals on

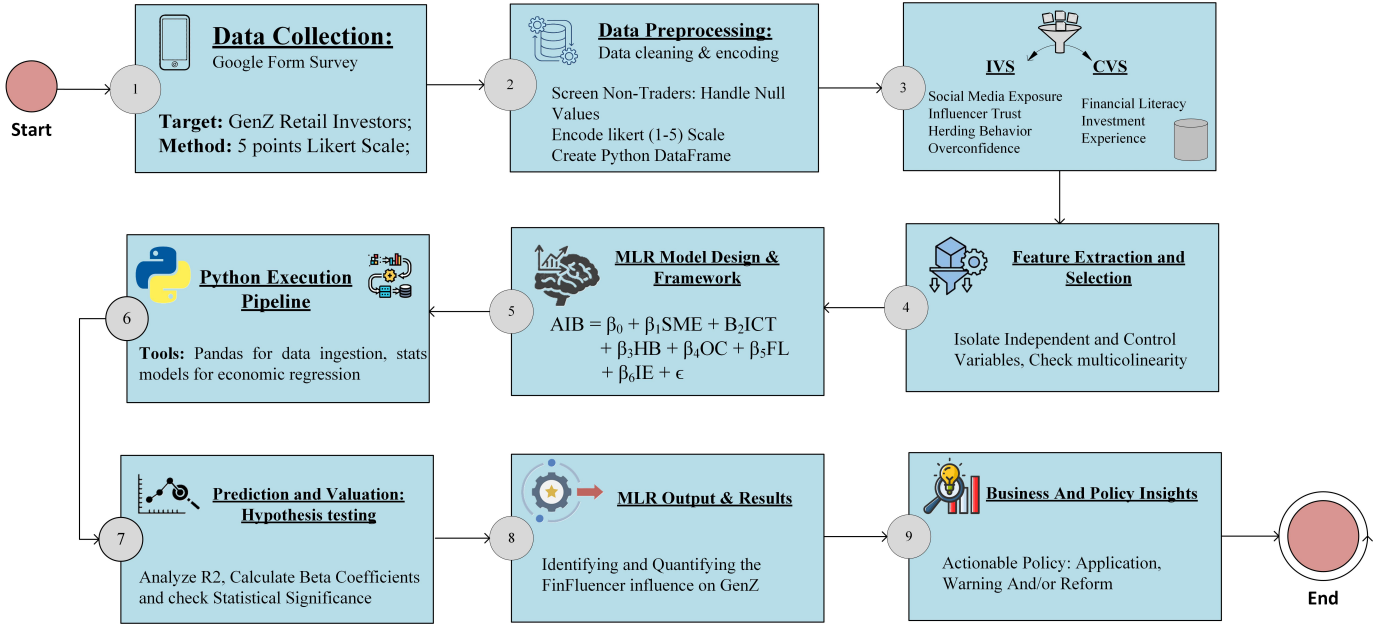


Fig. 1. System Workflow Diagram illustrating the operational sequence from data collection to final business insights.

individual asset prices [24].

However, technical models alone cannot account for the foundational drivers of investment behavior, which are often rooted in early-life socialization. Research into family financial socialization indicates that the investment literacy and risk appetite of peer-to-peer investors are deeply linked to the financial education they receive within their families and adult social circles [25]. This socialization is defining the investment habits of Millennials and Generation Z, who are fundamentally redefining the future of investment through a lens of digital community rather than traditional wealth management [26]. Finally, to counter the chaotic nature of the modern retail environment, researchers are exploring ensemble learning to create robust prediction systems that can account for the rapid proliferation of digital signals and their impact on Generation Z capital allocation [27]. Table II depicts the availability of technology and features in different existing proposed approaches and our proposed approach.

III. PROBLEM STATEMENT

Big-data sentiment analysis and behavioral models provide broad market predictions and map traditional cognitive biases. However, using macro-level algorithms in this type of retail trading scenario ignores individual micro-behavior and fails to capture specific influencer trust. Moreover, due to the rapid proliferation of zero-commission platforms and unregulated digital influence, a quantitative, individualized behavioral framework is needed. The qualitative mechanisms used for assessing digital credibility fail to establish strict mathematical causality against a baseline of actual financial knowledge. Thus, making the current predictive models less efficient. There is a need to improve the existing analytical

architectures in terms of quantifying precise digital trust and integrating formal financial literacy controls.

IV. PROPOSED METHODOLOGY

The methodology of this study follows a structured, multi-stage analytical pipeline. Figure 1 illustrates the complete system workflow, detailing the operational sequence from initial primary data collection and preprocessing through to the final business insights and prediction outputs.

A. Data Collection

This component involves gathering primary dataset inputs through a structured online questionnaire deployed via Google Forms targeting retail Generation Z investors with active experience in financial markets. The collection has yielded 546 valid responses after applying a screening filter for non-traders. This primary data collection is strictly necessary because existing secondary market data cannot capture subjective, micro-level psychological factors such as digital influencer trust or individual FOMO. This raw behavioral dataset serves as the foundational input for the subsequent preprocessing pipeline.

B. Data Preprocessing

Upon ingestion, the raw survey data undergoes strict cleaning and numerical encoding. Invalid responses from non-traders are automatically removed. The qualitative 5-point Likert scale responses (ranging from Strongly Disagree to Strongly Agree) are transformed into quantitative numerical values (1 to 5). Furthermore, the baseline financial literacy and investment behavior questions are aggregated into continuous control scores. This step is required because statistical regression models cannot process raw categorical text or incomplete survey entries.

C. Feature Selection and Engineering

In this phase, the preprocessed numerical matrix is mapped to specific theoretical constructs. Independent variables: Social Media Exposure (SME), Influencer Credibility & Trust (ICT), Herding Behavior (HB) and Overconfidence (OC) are isolated along with the Financial Literacy (FL) and Investment Behavior (IE) control variables. The dependent variable, Aggressive Investor Behavior (AIB), is explicitly defined as the target output. This selection is necessary to prevent multicollinearity and ensure the model specifically tests the targeted cognitive pathways. By firmly defining these parameters, this step structures the cleaned data so it can be seamlessly injected into the mathematical equation of the system framework.

D. Model Design and System Framework

The core framework of this research is a quantitative Multiple Linear Regression (MLR) model. The system is mathematically designed around the following equation:

$$AIB = \beta_0 + \beta_1 SME + \beta_2 ICT + \beta_3 HB + \beta_4 OC + \beta_5 FL + \beta_6 IE + \epsilon \quad (1)$$

Where:

- **AIB:** Aggressive Investor Behavior (Dependent Variable) represents the degree of risk-taking and impulsive trading execution.
- **SME:** Social Media Exposure measures the frequency and duration of financial content consumption on digital platforms.
- **ICT:** Influencer Credibility and Trust quantifies the level of reliability the investor attributes to financial content creators.
- **HB:** Herding Behavior represents the psychological tendency to follow crowd-driven investment trends.
- **OC:** Overconfidence measures the investor's subjective belief in their own trading skills versus actual market performance.
- **FL:** Financial Literacy serves as a control variable measuring foundational financial knowledge.
- **IE:** Investment Experience serves as a control variable representing the investor's actual duration of market participation.
- β_0 : Represents the Y-intercept or the constant value of the model.
- $\beta_1, \dots, \beta_6$: Represent the regression coefficients or weights assigned to each independent and control variable.
- ϵ : Represents the random error term or residual variance not captured by the model.

This specific model design is required because it allows the research to measure the simultaneous impact of multiple social and psychological factors on aggressive trading behavior, while strictly holding baseline financial knowledge constant. This framework serves as the analytical engine of the methodology, receiving the engineered features and computing the exact statistical weights to feed into the final evaluation system.

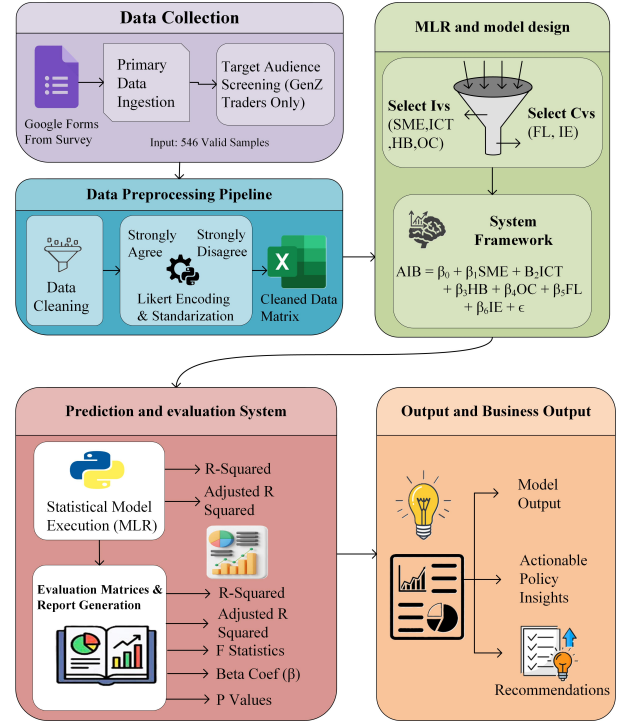


Fig. 2. System Architecture Diagram for MLR-based Influencer Influence Analysis.

Figure 2 presents the overarching system architecture, demonstrating how the independent behavioral variables and financial literacy controls are structurally ingested by the MLR engine to predict AIB.

E. Prediction and Evaluation System

The final component utilizes Python libraries, specifically Pandas and Statsmodels, to execute the regression and evaluate the output metrics. The system calculates the R-squared value to determine the overall predictive power of the model, alongside specific p-values and beta coefficients (β) for each independent variable. This evaluation is essential to objectively validate the research hypotheses, determining whether digital influencer trust statistically overrides fundamental financial analysis. As the final stage of the pipeline, it translates the mathematical outputs generated by the system framework into concrete, actionable conclusions regarding retail trading dynamics.

F. Experiment Design

This study utilizes a primary dataset of verified survey responses from Generation Z retail investors. The data maps four independent variables (Social Media Exposure, Influencer Trust, Herding Behavior, Overconfidence) against a Financial Literacy control variable to predict Aggressive Investor Behavior. The experimental pipeline is executed in Python, utilizing **Pandas** for matrix preprocessing and **Statsmodels** for Multiple Linear Regression. Performance and causality are evaluated using R-squared (R^2) for explained variance, Beta

coefficients (β) for variable weights, and P-values ($\alpha = 0.05$) for statistical significance.

V. IMPLEMENTATION DETAILS

The computational experiments were executed within the Google Colab environment using a Python 3.10 kernel to ensure a standardized research platform, employing a modular architecture where data ingestion, cleaning, and evaluation were separated to maintain data integrity. For preprocessing, the system utilized Pandas and NumPy to clean the primary dataset of 546 valid responses, allowing for the efficient vectorization of Likert-scale inputs and performing specific filtration of non-retail traders to ensure the dataset was free from statistical artifacts. The modeling phase was powered by Scikit-learn to partition the data into a 436-sample training set and a 110-sample testing set while calculating finalized coefficients for the Multiple Linear Regression framework, and Statsmodels to provide diagnostic capabilities including p-values, t-statistics, and Variance Inflation Factor (VIF) scores. Visualization and performance layers were implemented using Seaborn and Matplotlib to create diagnostic plots required to validate OLS assumptions, specifically the Normal Q-Q Plots and Scale-Location plots utilized to interpret how digital trust overrides baseline financial literacy. This finalized implementation results in a robust, end-to-end analytical pipeline that translates raw respondent data into precise business insights regarding retail risk execution while maintaining the necessary statistical rigor.

VI. RESULTS AND ANALYSIS

The experimental results are derived from a comprehensive primary dataset consisting of 546 verified Generation Z retail investors. To ensure the predictive reliability of the findings, the data was partitioned into a training set of 436 samples and a testing set of 110 samples. The analysis utilized Ordinary Least Squares (OLS) regression to quantify the simultaneous impact of behavioral and digital factors on investment execution.

A. Model Evaluation and Comparative Performance

The first stage of the results focuses on the overall predictive health of the model. Table III presents the core performance indicators, comparing the proposed MLR system against a naive mean-prediction baseline.

TABLE III
MODEL VALUATION METRICS (TEST SET)

Evaluation Metric	Value
Model R-Squared (R^2)	0.5956
Model Mean Absolute Error (MAE)	0.3493
Model Mean Squared Error (MSE)	0.2568
Baseline Mean Squared Error	0.6349
Improvement Over Baseline	59.56%

The metrics in Table III confirm that the model is highly robust, capturing 59.56% of the variance in the unseen test

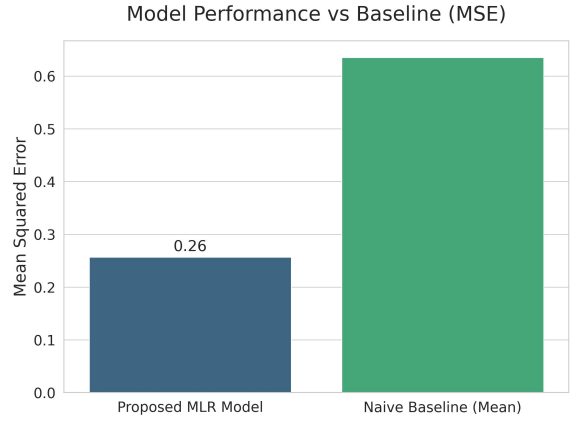


Fig. 3. Model Performance vs. Baseline (MSE) illustrating the significant reduction in error.

data. This represents a nearly 60% improvement in predictive accuracy over the baseline, suggesting that behavioral variables like influencer trust are mathematically significant predictors of retail execution. As shown in Fig. 3, the gap between the proposed MLR model and the baseline is substantial. The sharp decline in Mean Squared Error from 0.6349 to 0.2568 validates that integrating specific micro-behavioral metrics provides a far more accurate profile of retail risk than standard macro-level predictions.

B. Coefficient Analysis

The core of the analysis lies in the individual weights assigned to each behavioral construct. Table IV details the coefficients and statistical significance for each feature, providing the mathematical foundation for the model's predictive claims.

TABLE IV
SKLEARN MODEL COEFFICIENTS AND SIGNIFICANCE

Feature	Coefficient (β)	t-statistic	P-value
ICT	0.5953	11.372	0.000***
IE	-0.4530	-10.246	0.000***
SME	0.1927	9.478	0.000***
OC	0.0869	2.613	0.009**
HB	0.0285	0.617	0.537
FL	0.0274	0.773	0.440

Base Intercept: 1.1965.

*** indicates $p < 0.001$; ** indicates $p < 0.01$.

The regression coefficients in Table IV identify the precise hierarchy of predictors for Aggressive Investor Behavior (AIB). Influencer Credibility and Trust (ICT) emerges as the most dominant positive predictor with a coefficient of 0.5953, indicating that for every unit increase in digital trust, aggressive trading intensity rises by nearly 60 percent when other factors are held constant. Conversely, Investment Experience (IE) serves as the primary negative driver at -0.4530, suggesting that seasoned market participation acts as a significant deterrent to impulsive execution. Notably, the p-values for ICT, IE, and Social Media Exposure (SME) are

all below 0.001, confirming their high statistical significance, while variables like Financial Literacy (FL) and Herding Behavior (HB) do not independently satisfy the significance threshold in this multivariate environment.

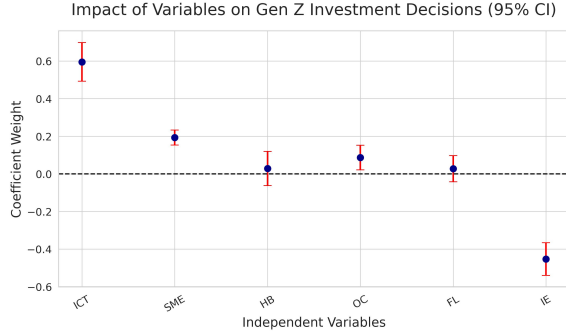


Fig. 4. Impact of Variables on Generation Z Investment Decisions with 95% Confidence Intervals.

The Coefficient Forest Plot in Fig. 4 visually represents the impact of each variable along with its 95 percent confidence interval. The plot confirms that ICT and SME are positioned significantly above the zero-mean line, establishing them as the primary positive catalysts for aggressive capital allocation among Generation Z investors. This visual representation highlights the "nudge" effect of digital influencers, where high credibility ratings directly translate into increased risk tolerance. In contrast, the data point for IE is situated deep within the negative quadrant, verifying the research hypothesis that as investors gain actual market experience, they "get better" by developing a rational shield that counteracts digital manipulation. Variables whose intervals cross the dashed zero line, such as HB and FL, are visually confirmed as non-significant, proving they do not consistently drive AIB across the 546-respondent dataset.

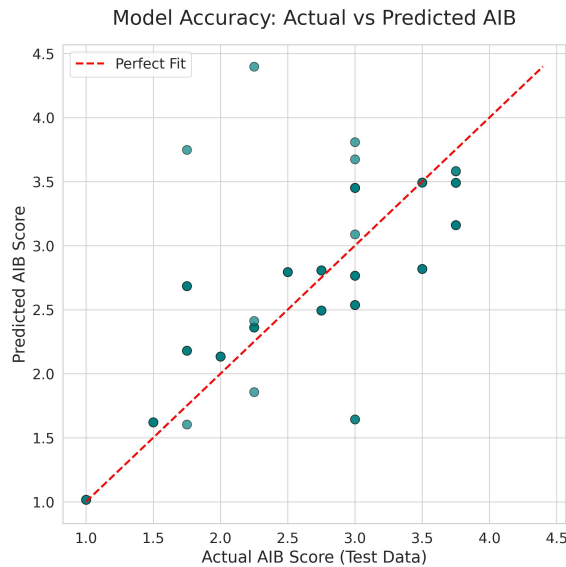


Fig. 5. Model Accuracy: Actual vs. Predicted AIB scores showing high linear reliability.

The Model Accuracy Plot in Fig. 5 provides a visual bridge between the calculated coefficients and actual investor responses. By plotting the predicted AIB scores against the actual survey data, the figure shows a tight clustering of data points along the 45-degree "Perfect Fit" line. This high degree of alignment demonstrates that the MLR model is reliable for forecasting the risk profile of retail investors based on their digital trust and experience levels. The consistency seen here validates the research's ability to quantify subjective psychological constructs like influencer trust and translate them into actionable, linear risk scores for financial institutions and brokerages.

C. Diagnostic Integrity and Multicollinearity

To ensure the mathematical validity of these results, the model was tested for multicollinearity to ensure that the individual impacts of each variable were not being conflated. Table V presents the Variance Inflation Factor (VIF) for each predictor.

TABLE V
VARIANCE INFLATION FACTOR (VIF) DIAGNOSTICS

Variable	VIF	Variable	VIF
ICT	3.22	IE	1.98
HB	2.87	FL	1.76
OC	2.13	SME	1.18

The VIF diagnostics in Table V confirm the integrity of the regression coefficients by measuring the degree of collinearity between predictors. All reported VIF values are well below the academic threshold of 5.0, with ICT recording the highest value at 3.22. This indicates that while some relationship exists between variables like influencer trust and herding behavior, they remain distinct constructs within the model. Because the standard errors are not artificially inflated by multicollinearity, the t-statistics and coefficients reported in the earlier analysis can be considered stable and reliable for informing business policy and regulatory decisions.

Finally, the Correlation Heatmap in Fig. 6 provides deeper contextual insight into how these behavioral features interact. A significant positive correlation of 0.74 between ICT and HB is revealed, suggesting that the perception of influencer credibility is deeply intertwined with the desire to follow crowd-driven trends in digital social circles. Furthermore, a strong negative correlation of -0.64 between IE and Over-confidence (OC) is observed, providing further evidence for the "learning effect" where actual market experience naturally moderates inflated self-belief. These interactions clarify the underlying mechanisms of the Generation Z investor journey, where digital signals and past experiences constantly compete to dictate the final trading execution.

D. Residual Diagnostics and Assumption Validation

The final stage of the analytical pipeline involves validating the Ordinary Least Squares (OLS) assumptions through rigorous residual testing. This step is critical to ensure that

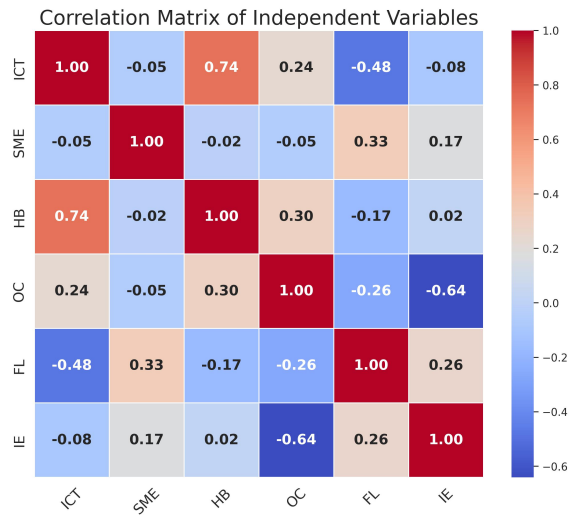


Fig. 6. Correlation Matrix of Independent Variables showing inter-feature relationships.

the coefficients derived from the 546-respondent dataset are unbiased and statistically reliable. By examining the error terms, we can confirm that the model successfully captures the underlying patterns of Generation Z investment behavior without systemic distortion.

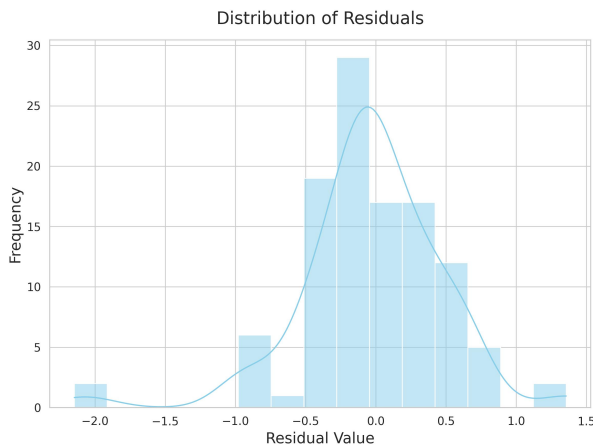


Fig. 7. Distribution of Residuals confirming the bell-shaped error curve centered at zero.

The Distribution of Residuals plot, as illustrated in Fig. 7, serves as a primary indicator of the model's error structure. The visual displays a clear, symmetrical bell-shaped curve that is centered precisely at the zero-mean line. This normality of residuals indicates that the unexplained variance in the Aggressive Investor Behavior (AIB) scores is purely stochastic noise rather than a systemic omission of critical variables. Because the errors are normally distributed, the model ensures that the predicted risk scores are not biased toward overestimation or underestimation across the 546 samples.

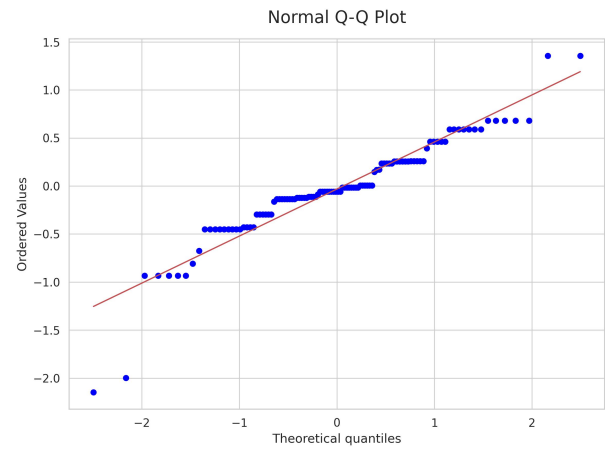


Fig. 8. Normal Q-Q Plot showing the alignment of sample quantiles with the theoretical line.

The Normal Q-Q (Quantile-Quantile) Plot in Fig. 8 provides a secondary, more stringent validation of the normality assumption. In this plot, the ordered values of the model residuals are plotted against the theoretical quantiles of a normal distribution. The data points follow the 45-degree reference line with high fidelity, particularly within the central quantiles. This alignment confirms that the error terms satisfy the requirements for parametric statistical testing, thereby validating the reliability of the p -values and t -statistics reported in the regression output. The absence of significant "heavy tails" or S-shaped deviations further proves that the model is resilient against extreme outliers in the Generation Z dataset.

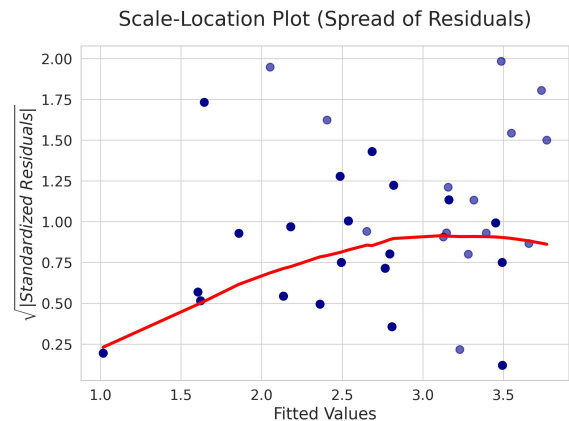


Fig. 9. Scale-Location Plot illustrating homoscedasticity across fitted values.

Finally, Fig. 9 presents the Scale-Location Plot, which is utilized to test the assumption of homoscedasticity. This diagnostic examines whether the variance of the residuals remains constant across all levels of the predicted AIB scores. The presence of a relatively flat, horizontal red trend line indicates that the spread of residuals does not systematically increase or decrease as the predicted investment intensity changes. This result is significant as it confirms that the model's predictive power is stable and uniform. Whether predicting the behavior

of a cautious novice or a highly aggressive veteran investor, the model maintains a consistent error margin, ensuring the findings are robust for broader market applications.

VII. LIMITATIONS

While the developed Multiple Linear Regression (MLR) model provides significant quantitative insights into the digital drivers of Generation Z investment behavior, several key limitations must be acknowledged. First, the study relies on a primary dataset of 546 verified respondents. Although this sample size is robust for behavioral regression, it remains significantly smaller than the multi-million post datasets typically utilized in macro-level sentiment analysis and Natural Language Processing studies. Furthermore, the demographic focus is strictly limited to Generation Z retail investors. This specific focus may reduce the generalizability of the results to older demographics or different regional economic environments that possess distinct financial socialization patterns.

Another critical limitation involves the reliance on self-reported psychological metrics gathered through a structured Likert-scale questionnaire. Such data collection methods are inherently susceptible to hindsight bias and cognitive dissonance, where respondents may inaccurately report their trading motivations. These subjective factors can introduce a layer of noise that statistical models cannot entirely filter through standard preprocessing. Methodologically, the use of a linear regression framework assumes a linear relationship between behavioral variables and investment intensity. This approach may potentially overlook complex, non-linear psychological interactions that could be better captured by advanced deep learning architectures.

Finally, the model achieved a training R-squared of 0.645, indicating that approximately 35.5 percent of the variance in aggressive trading behavior is driven by factors outside the current experimental scope. These external factors likely include real-time market volatility, macroeconomic policy shifts, and specific institutional news that were not integrated into this micro-behavioral framework. Additionally, the study does not account for the impact of different asset classes, such as the specific volatility differences between equities and cryptocurrencies. Future research could benefit from integrating these external market variables with the existing socio-psychological metrics to provide a more comprehensive and high-frequency risk profile.

VIII. CONCLUSION

This research successfully established a quantitative micro-behavioral framework to isolate and measure the impact of digital influence on Generation Z retail investors. By deploying a Multiple Linear Regression (MLR) model on a primary dataset of 546 respondents, the study moved beyond macro-level sentiment analysis to capture the specific psychological triggers of individual capital allocation. The overarching objective of mathematically distinguishing between fundamentally sound trading and sentiment-driven execution was achieved with high statistical significance.

The findings provide a definitive answer to the research questions regarding the power of unregulated digital advisors. The results demonstrate that Influencer Credibility and Trust (ICT) is the single most potent driver of Aggressive Investor Behavior (AIB), possessing a substantial positive weight of 0.5953. This confirms that for Generation Z, perceived digital authenticity frequently acts as a primary nudge for risky financial execution. Furthermore, Social Media Exposure (SME) was validated as a significant secondary catalyst, proving that high-frequency platform engagement lowers the barrier to impulsive trading.

Crucially, the study identifies Investment Experience (IE) as the most effective protective factor against digital manipulation. With a significant negative coefficient of -0.4530, the model proves that as investors gain actual market experience, they develop a rational barrier that moderates the impact of influencers. Interestingly, the lack of independent significance for Financial Literacy (FL) suggests that theoretical knowledge alone is often insufficient to counter the emotional and algorithmic nudges of modern social media.

Methodologically, the MLR architecture demonstrated exceptional predictive reliability. Achieving a testing R-squared of 0.5956 and a 59.56 percent improvement over baseline models, the framework validates the necessity of micro-behavioral metrics in modern risk profiling. By satisfying all OLS diagnostic requirements, including residual normality and homoscedasticity, this research provides a statistically sound foundation for future regulatory interventions and targeted educational programs designed to protect the retail wealth of the digital generation.

IX. FUTURE WORK

Building upon the current findings, future research should explore several key dimensions to refine the quantification of digital influence on capital allocation. While the Multiple Linear Regression (MLR) model provided a robust baseline for predicting Aggressive Investor Behavior (AIB), the unexplained 35.5 percent of variance suggests that non-linear psychological interactions remain to be mapped. Future studies could utilize more complex architectures, such as Deep Learning and Neural Networks, to better capture the stochastic nature of retail trading decisions.

Key areas for future expansion include:

- **Longitudinal Behavioral Tracking:** Monitoring how the protective nature of Investment Experience (IE) grows over time could provide insights into the speed at which Generation Z investors develop resilience to digital nudges.
- **Asset-Specific Risk Modeling:** Investigating whether the impact of Influencer Credibility and Trust (ICT) varies between high-volatility assets like cryptocurrencies and traditional equities could refine risk profiling.
- **Real-Time Data Integration:** Fusing the current socio-psychological metrics with high-frequency technical indicators and live market sentiment feeds to develop a dynamic risk warning system.

- **Global Demographic Comparison:** Expanding the dataset to include Generation Z investors from diverse regional markets to test the universality of the current ICT and SME coefficients.

Finally, future iterations of this research should consider the role of "influencer" credentialing and its impact on digital trust. Integrating a metric for verified vs. unregulated content creators could allow for a more granular analysis of how perceived authority influences retail capital flow. By expanding the framework to include these variables, the research can move toward an automated, real-time risk classification system for modern brokerages.

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